Suppose the true model is

$$y = a + \beta x + \gamma z + \varepsilon, \tag{1}$$

where x is the variable of interest, z is a control variable.

The relationship between x and z is

$$z = \alpha + \pi x + u \tag{2}$$

If z is omitted in the regression, you are estimating

$$y = a + \beta x + \gamma(\alpha + \pi x + u) + \varepsilon \tag{3}$$

$$= (a + \gamma\alpha) + (\beta + \gamma\pi)x + (\varepsilon + \gamma u) \tag{4}$$

$$= \widehat{const}_{bias} + \widehat{\beta}_{bias} x + \widehat{\varepsilon}_{bias}. \tag{5}$$

The biased estimate of β is

$$\widehat{\beta}_{bias} = \beta + \gamma\pi, \tag{6}$$

where β indicates the true relationship between y and x, γ is the true relationship between y and z, π is the true relationship between x and z.

The Table 1 below reflects the change in the estimate of β if z is omitted from regressions.

Table 1

	$\gamma > 0$	$\gamma < 0$
$\pi > 0$	$\widehat{\beta}_{bias} > \beta$	$\widehat{\beta}_{bias} < \beta$
$\pi < 0$	$\widehat{\beta}_{bias} < \beta$	$\widehat{\beta}_{bias} > \beta$